CHAPTER 3

TRIGONOMETRIC FUNCTIONS

CHAPTER ESSENCE AT A GLANCE:
Trigonometry rules the circle! Angles in standard position turn counter-clockwise. The ASTC rule ('All Silver Tea Cups') tells you in 2 seconds whether Sin, Cos, or Tan is positive in any of the 4 quadrants. Master the double-angle formulas for guaranteed exam points!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Degree & Radian Relation:** π radians = 180° implies 1 radian = 180° over π. Arc length formula: l = r θ (θ MUST be in radians!).
- **ASTC Quadrant Rule:** Quadrant 1: All positive. Quadrant 2: Sin & Cosec positive. Quadrant 3: Tan & Cot positive. Quadrant 4: Cos & Sec positive.
- **Even & Odd Trig Functions:** cos(-x) = cos x and sec(-x) = sec x (Even, minus disappears!). sin(-x) = -sin x and tan(-x) = -tan x (Odd, minus pulls out!).
- **Compound Angle Formulas:** sin(A ± B) = sin A cos B ± cos A sin B ; cos(A ± B) = cos A cos B -+ sin A sin B.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$l = r\theta \quad (\theta \text{ in radians}), \quad 1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57^\circ 16'$$

Arc Length and Radian-Degree Conversion Relation

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y, \quad \cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

Fundamental Compound Angle Addition Theorems

$$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

Double Angle Cosine Four-Fold Expansions

$$\sin x + \sin y = 2\sin \frac{x+y}{2} \cos \frac{x-y}{2}, \quad \cos x + \cos y = 2\cos \frac{x+y}{2} \cos \frac{x-y}{2}$$

Sum-to-Product Transformation Identities (CD Formulas)

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** cos(-x) = cos x (Even function, minus disappears!), whereas sin(-x) = -sin x and tan(-x) = -tan x (Odd functions, minus pulls out!).
- **Alert:** Convert degrees to radians by multiplying by π over 180. E.g., 30° = π over 6, 45° = π over 4, 60° = π over 3, 90° = π over 2!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the values of: (a) sin(765 deg) (b) tan(19 pi over 3). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: sin(765°)]** 765 = 2 × 360 + 45°. sin(765°) = sin(45°) = 1 over √2. [1 Mark]

• **[Step 2: tan(19 π over 3)]** 19 π over 3 = 6 π + π over 3. tan(6 π + π over 3) = tan(π over 3) = √3. [1 Mark]

• **[Boxed Result: (a) 1 over √2, (b) √3]**

2. Find the value of sin(15 deg) and cos(75 deg). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: sin(15°)]** sin(45° - 30°) = sin 45 cos 30 - cos 45 sin 30 = (1 over √2)(√3 over 2) - (1 over √2)(1 over 2) = (√3 - 1) over (2 √2). [1 Mark]

• **[Step 2: cos(75°)]** cos(45° + 30°) = cos 45 cos 30 - sin 45 sin 30 = (√3 - 1) over (2 √2). [1 Mark]

• **[Boxed Result: sin 15° = cos 75° = (√3 - 1) over (2 √2)]**

3. Prove that: cos(pi over 4 - x) cos(pi over 4 - y) - sin(pi over 4 - x) sin(pi over 4 - y) = sin(x + y). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Compound Angle Identity]** Identity: cos A cos B - sin A sin B = cos(A + B). Let A = π over 4 - x and B = π over 4 - y. [1 Mark]

• **[Step 2: Evaluation]** LHS = cos[(π over 4 - x) + (π over 4 - y)] = cos[π over 2 - (x + y)]. [1 Mark]

• **[Step 3: Co-function Identity]** Since cos(π over 2 - θ) = sin θ: cos[π over 2 - (x + y)] = sin(x + y) = RHS. Hence proved. [1 Mark]

• **[Boxed Result: Identity proved: LHS = sin(x + y)]**

4. Prove that: [tan(pi over 4 + x)] over [tan(pi over 4 - x)] = [(1 + tan x) over (1 - tan x)]^2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Expansion of tan(A + B)]** tan(π over 4 + x) = (tan π over 4 + tan x) over (1 - tan π over 4 tan x) = (1 + tan x) over (1 - tan x). [1 Mark]

• **[Step 2: Expansion of tan(A - B)]** tan(π over 4 - x) = (tan π over 4 - tan x) over (1 + tan π over 4 tan x) = (1 - tan x) over (1 + tan x). [1 Mark]

• **[Step 3: Dividing]** LHS = [(1 + tan x) over (1 - tan x)] over [(1 - tan x) over (1 + tan x)] = [(1 + tan x) over (1 - tan x)]^2 = RHS. Hence proved. [1 Mark]

• **[Boxed Result: Identity proved: [(1 + tan x) over (1 - tan x)]^2]**

5. Prove that: cos(4x) = 1 - 8 sin^2(x) cos^2(x). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Double Angle Identity]** cos(4x) = cos(2(2x)) = 1 - 2 sin^2(2x). [1.5 Marks]

• **[Step 2: Expansion of sin(2x)]** Since sin(2x) = 2 sin x cos x: sin^2(2x) = (2 sin x cos x)^2 = 4 sin^2 x cos^2 x. Substituting: cos(4x) = 1 - 2 (4 sin^2 x cos^2 x) = 1 - 8 sin^2 x cos^2 x = RHS. Hence proved. [1.5 Marks]

• **[Boxed Result: cos 4x = 1 - 8 sin^2 x cos^2 x proved]**

6. If cot x = -5 over 12 and x lies in the second quadrant, find the values of all other five trigonometric functions. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: tan x and cosec x]** In Quadrant II: sin x and cosec x are positive; cos x, sec x, tan x, cot x are negative. tan x = 1 over cot x = -12 over 5. [1 Mark]

• **[Step 2: cosec x and sin x]** cosec^2 x = 1 + cot^2 x = 1 + (-5 over 12)^2 = 1 + 25 over 144 = 169 over 144 ⇒ cosec x = +13 over 12. sin x = 1 over cosec x = 12 over 13. [1.5 Marks]

• **[Step 3: cos x and sec x]** cos x = cot x . sin x = (-5 over 12) x (12 over 13) = -5 over 13. sec x = 1 over cos x = -13 over 5. [1.5 Marks]

• **[Boxed Result: sin x = 12 over 13, cos x = -5 over 13, tan x = -12 over 5, cosec x = 13 over 12, sec x = -13 over 5]**

7. Prove that: (cos 4x + cos 3x + cos 2x) over (sin 4x + sin 3x + sin 2x) = cot 3x. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Grouping Extremes]** Numerator = (cos 4x + cos 2x) + cos 3x. Denominator = (sin 4x + sin 2x) + sin 3x. [1 Mark]

• **[Step 2: Applying C-D Formulas]** cos 4x + cos 2x = 2 cos 3x cos x. Numerator = 2 cos 3x cos x + cos 3x = cos 3x (2 cos x + 1). [1.5 Marks]

• **[Step 3: Denominator Factoring]** sin 4x + sin 2x = 2 sin 3x cos x. Denominator = 2 sin 3x cos x + sin 3x = sin 3x (2 cos x + 1). LHS = [cos 3x (2 cos x + 1)] over [sin 3x (2 cos x + 1)] = cos 3x over sin 3x = cot 3x = RHS. Hence proved. [1.5 Marks]

• **[Boxed Result: Identity proved: cot 3x]**